

Rangesh Sashta Vasudevan

+1 (313) 210-2503 | rangeshsasthavasudevan@gmail.com | Detroit, MI | [LinkedIn](#)

PROFESSIONAL SUMMARY

Robotics Engineer specializing in humanoid and autonomous robotic systems, with an M.S. in Robotics – Intelligent Control from Wayne State University and 3+ years of hands-on experience spanning manipulator design, mobile robot navigation, and human-robot interaction. Proven ability to deliver end-to-end robotic systems — from kinematic modeling and motion planning through ROS2 software integration and hardware validation — in both research and industry environments. Co-authored 2 peer-reviewed publications; 1 M.S. thesis in submission.

WORK EXPERIENCE

Research Assistant – Robotics & Intelligent Systems

Jan 2025 – May 2026

CARES Lab (Computer Assisted Robot Enhanced Systems), Wayne State University

Detroit, MI

- Engineered real-time operator-state monitoring pipeline using multimodal sensor fusion (EEG, IMU, video), reducing teleoperation error events by enabling proactive safety interventions in robotic surgery scenarios.
- Designed and deployed ML classification models (Random Forest, K-Means) on EEG time-series data achieving >85% cognitive load detection accuracy across 20+ test sessions.
- Built full data acquisition and feature extraction pipeline integrating Muse 2 EEG headband with TurtleBot3 teleoperation interface for real-time human-robot interaction studies.
- Contributed to 2 peer-reviewed publications and 1 in-submission M.S. thesis on intelligent control and operator safety in teleoperated robotic systems.

Robotics Engineer

May 2023 – Dec 2023

Gritsys Technologies Pvt. Ltd

Chennai, India

- Designed, developed, and tested humanoid robotic systems for healthcare and assistive applications, integrating 6-DOF manipulator arms with custom end-effectors for patient-assist tasks.
- Integrated IMU, force/torque, and proximity sensors with embedded control systems, improving robot responsiveness and joint-level compliance for safe human-robot interaction.
- Applied forward/inverse kinematic modeling and trajectory planning algorithms to optimize motion profiles, reducing cycle time by ~15% on repetitive assist tasks.
- Programmed, debugged, and validated robotic software and hardware components across full system stack — from embedded firmware to high-level ROS motion planning nodes.

Robotics Engineering Intern

Jun 2022 – Apr 2023

Gritsys Technologies Pvt. Ltd

Chennai, India

- Assisted in design, development, and testing of a humanoid robot for healthcare applications, gaining hands-on experience across mechanical assembly, control programming, and system debugging.
- Implemented sensor fusion routines combining IMU and encoder data to improve robot joint state estimation and motion smoothness.
- Explored robotic kinematics, obstacle avoidance, and introductory ML methods to enhance robot perception and motion in unstructured environments.

Research Assistant – Soft Robotics & Flexible Electronics

Mar 2022 – Apr 2023

Center for Additive Manufacturing, Chennai Institute of Technology

Chennai, India

- Designed and fabricated soft robotic grippers and latch mechanisms using flexible printed circuit techniques, targeting compliance and adaptability in unstructured manipulation tasks.
- Led experimental setup, mechanical testing, and technical documentation, contributing findings to 1 peer-reviewed journal publication (IOP Publishing, 2025)..
- Investigated novel materials and additive manufacturing methods to optimize soft actuator performance for real-world robotic applications.

EDUCATION

Wayne State University

Jan 2024 – May 2026

M.S. Robotics – Intelligent Control | GPA: 4.00 / 4.00

Chennai Institute of Technology

Jun 2019 – Apr 2023

B.E. Mechatronics Engineering | GPA: 8.84 / 10.00

PROJECTS

Real-Time Cognitive Load Detection for Teleoperated Robotic Surgery – M.S. Thesis

Jan 2025 – May 2026

Tools: Python, Machine Learning, Signal Processing, Muse 2 EEG, ROS2

- Developed real-time cognitive load detection framework using EEG signals from Muse 2 headband, integrated with a surgical teleoperation interface to enable closed-loop operator safety monitoring.
- Built end-to-end signal processing pipeline: artifact removal → spectral feature extraction (alpha/beta/theta bands) → ML classification (Random Forest, K-Means), achieving >85% classification accuracy.
- Evaluated framework across teleoperation safety, fatigue detection, and HRI scenarios; results submitted as M.S. thesis to Wayne State ECE Dept. (2026).

Autonomous Mobile Robot Navigation & Operator Safety System

Aug 2025 – Dec 2025

Tools: ROS2, Python, TurtleBot3, OpenCV, LiDAR, AprilTag, SLAM, Nav2

- Developed full ROS2-based autonomous mobile robot stack: SLAM, Nav2 autonomous navigation, LiDAR obstacle avoidance, and joystick teleoperation on TurtleBot3 platform.
- Integrated AprilTag fiducial tracking and OpenCV-based computer vision for real-time localization and environmental awareness in dynamic environments.
- Fused EEG fatigue monitoring with robot control loop to automatically trigger safety protocols (speed reduction, stop) when operator fatigue exceeded threshold — novel HRI safety architecture.

3-DOF Robotic Manipulator with Kinematic Modeling & ROS Integration

Aug 2024 – Dec 2024

Tools: MATLAB, ROS, Python, Path Planning, Optimization Algorithms, GUI

- Designed 3-axis robotic manipulator with full kinematic model; implemented forward and inverse kinematics with MATLAB Robotics Toolbox and custom GUI for real-time joint control.
- Integrated ROS motion planning and path-planning algorithms (A*, RRT) for autonomous task execution with AI-assisted trajectory optimization.

Pneumatic Soft Gripper for KUKA KR16 Industrial Manipulator

Jan 2023 – Apr 2023

Tools: Soft Robotics, KUKA KR16, Pneumatics, FEA, KUKA.Sim, Fusion 360

- Designed and fabricated pneumatic soft robotic gripper for KUKA KR16 industrial manipulator; performed FEA to optimize compliance, grip force, and durability across pick-and-place cycles.
- Programmed KUKA KR16 using teach pendant and KUKA.Sim for simulation, trajectory validation, and end-effector integration testing in industrial lab environment.

Warrior-1 Autonomous Ground Vehicle – IGVC 2026 (Suspension Lead)

Aug 2025 – May 2026

Tools: Fusion 360, ANSYS Mechanical, Structural Analysis, Suspension Design

- Led mechanical design of double-wishbone suspension system for Wayne State IGVC autonomous ground vehicle; developed full CAD assemblies in Fusion 360 and performed static structural analysis in ANSYS to validate load capacity and system integrity.

PUBLICATIONS

- [C.1] Rangesh Sashta Vasudevan. (2023). [Design and Analysis of Latches Using RHBD Technology](#). 2023 9th International Conference on Smart Structures and Systems (ICSSS), IEEE. DOI: 10.1109/ICSSS58085.2023.10407965
- [C.2] Rangesh Sashta Vasudevan. (2025). [A Critical Review on Printed Electronics and Its Application](#). *Nanotechnology*, Vol. 36, No. 16, IOP Publishing. DOI 10.1088/1361-6528/adbc4
- [S.1] Rangesh Sashta Vasudevan. Real-Time Cognitive Load Detection Using EEG and Machine Learning in Robotic Surgeries. M.S. Thesis, Dept. of ECE, Wayne State University, 2026. (In Submission)

SKILLS

Robotics & Autonomous Systems: ROS2 • SLAM • Nav2 • Motion Planning • Path Planning • Trajectory Planning • Autonomous Navigation • Robot Kinematics & Dynamics • Robotic Manipulation • End-Effector Design • Humanoid Robotics • Mobile Robots • Teleoperation • Soft Robotics • Industrial Robotics • KUKA KR16 • PLC logic fundamentals • AprilTag Navigation

Programming: Python • C++ • MATLAB • Linux (Ubuntu) • Embedded Systems • ROS2 Development

AI, Perception & Control: Machine Learning • Deep Learning • Computer Vision • OpenCV • Signal Processing • Sensor Fusion • Intelligent Control Systems • Feature Extraction • EEG Signal Analysis • Cognitive Load Detection • Behavioral Signal Analysis

Simulation & CAD Tools: Gazebo • RViz • CARLA Simulator • MATLAB Robotics Toolbox • Simulink • ANSYS Mechanical • FEA • Fusion 360 • KUKA.Sim

Hardware & Integration: Sensor Integration • LiDAR • Camera Systems • Embedded Control • Pneumatic Actuation • Electro-Pneumatic Systems • Multimodal Sensor Fusion • Force/Torque Sensors